# Lecture 3

Evaluation and auditing: what does a score establish?

AI-assisted reconstruction from course slides, not a lecture transcript. Discussion reasoning is inferred; classroom responses were not supplied.

**The lesson.** An evaluation supports a claim only for its population, conditions, access rules, and measurement procedure. Auditing adds an assessment against expectations and a route to accountability. The lecture moves from generalization to subgroup errors, distribution shift, contaminated benchmarks, LLM judges, and the institutions that can act on findings.

## Match the evaluation design to the deployment claim

The No Free Lunch statement assumes all binary labelings of a finite input space are equally likely; expected error on unseen inputs is then 50%. This does not imply that structured real tasks are unlearnable.

An i.i.d. test estimates performance when train and test draws share a distribution. Deployment may violate that premise. The diagrams distinguish changes in input distribution (*covariate shift*, with $P(Y \mid X)$ unchanged), corrupted labels, changes in the feature–outcome relationship (*concept shift*), and changes in subgroup proportions. Different shifts call for different evidence.

The WILDS examples make the unit of generalization concrete: new camera locations, hospitals, times, or regions. Training at hospitals A–C, validating at D, and testing at E tests transfer to a held-out site. Randomly distributing images from all hospitals across splits mainly tests new images from already represented sites. Neither split answers every deployment question; choose according to the claim.

Reweighting changes the influence of observed cases and can target a different mixture when suitable support and assumptions hold. It cannot supply missing kinds of examples. The deep-learning illustration also cautions that changing weights need not change the final decision boundary as expected throughout training.

## Gender Shades: average performance can hide the central failure

The external study built the Pilot Parliaments Benchmark: 1,270 public images from six countries, evaluated by skin-type and perceived-gender groups. In the original results, darker-female subgroup errors were about 21–35%, while lighter-male errors were at most 0.8%. The intersection reveals a disparity that an overall average, or a single coarse breakdown, can obscure.

The follow-up reported substantial improvements, with some disparities remaining. This shows why external evaluation and public reporting can matter; it does not establish that every audit causes improvement or that better classification settles whether classifying gender from faces is an appropriate task. Keep the study's labels, population, and normative question separate.

## Prevent information leakage at the relevant boundary

Fit preprocessing on training data; use validation for development decisions; use the test set after those decisions are fixed. Separate related records by person, site, or time when needed for the claim. Patient overlap, post-outcome features, and test-informed preprocessing admit information unavailable in the intended prediction setting. A nominally separate test file is insufficient.

### A benchmark can be compromised through several routes

**Training exposure:** benchmark examples enter pretraining, undermining held-out measurement. **Selection exposure:** repeated model or prompt selection exploits benchmark feedback. **Evaluation-time access:** a tool-using agent retrieves answers or reference solutions. These are different mechanisms; an evaluation must specify the information and actions allowed at each stage. Web-scale training and empirical scaling curves do not resolve provenance.

The slides' answer-seeking incident illustrates the last route: a task score can reflect information access as well as the intended capability. Agent evaluation must therefore examine behaviour and access boundaries alongside answer correctness.

### Open-ended outputs require an explicit judgement procedure

For summaries or dialogue there may be many acceptable outputs. Specify factual accuracy, coverage, relevance, clarity, or harms; identify whose judgement is being represented and for which use. A domain expert, user, annotator, and LLM judge need not measure the same thing.

LLM judging provides scale, but needs validation. In the position-bias example, swapping two fixed answers changes which wins. Order balancing or repeated comparisons can reveal this sensitivity; agreement alone does not validate criteria.

Chatbot Arena measures preferences under its prompts, users, and comparison process. The "Leaderboard Illusion" discussion raises selection and opportunity concerns: more submitted variants and comparisons can increase the best observed score. The slides also give Arena's response disputing preferential access. This methodological dispute does not establish that the association is causal.

### When does evaluation become an audit?

Evaluation measures behaviour under specified conditions. An audit systematically assesses a specified system or process against stated expectations, with sufficient independence to support accountability. Subgroup error measurement can form part of an audit; it needs scope, criteria, evidence, reporting, and responsible actors to support a broader conclusion and corrective action.

Internal auditors can access development records and intervene early, but need freedom to challenge developers. External auditors can question the organization's framing, but may lack access. **Access, independence, and authority are separate properties.** Funding, publication rights, conflicts of interest, technical competence, affected-community participation, and powers to require change determine whether findings can matter. Embedded evaluators gain access; that alone does not establish independence or enforcement power.

A response to "who should audit?" should allocate roles: affected people identify problems; evaluators design tests; auditors assess evidence; decision makers require changes; oversight bodies review compliance. Specify who chooses the standard and can publish an unwelcome finding.

**Midterm emphasis (inference).** Practise diagnosing leakage and shift, designing the right split, explaining an intersectional audit result, and identifying what evidence and authority an audit lacks. Judge bias and benchmark exposure are strong conceptual candidates. Recalling leaderboard ranks or news dates has less support as a study priority than explaining the failure mechanism.

**Slide trail.** L03.pdf, pp. 3–8 (generalization/Gender Shades), 9–17 (shift and hygiene), 18–25 (benchmark exposure and judges), 26–32 (auditing); annotated copy consulted. Page numbers refer to PDF pages, not the slides' continuous printed numbering.